Part of SOW

Always External **Input** GPIO → FIN1 (PC5) and TRP1 (PE4)
(FIN and TRP signals are active LOW and generated through external buttons)

Always Output GPIO → FOUT1 (PC4)

GPIO Function

Default state of Input and Output Pins → HIGH

If FIN received (LOW) then make FOUT(LOW) till the FIN continues to remain (LOW)

If FIN returns (HIGH) → Again FOUT in Default state (HIGH)

Interrupt & Timer Function

When FIN received (LOW) >> After few microsecond TRP is also received (LOW) externally.

(TRP is 5 *uS* pulse triggered by mechanical system of equipment when set limits of operation is achieved. TRP must be associated with Interrupt Function).

When TRP received LOW then FIN will automatically be released to Default State (HIGH) Externally because equipment is shut after set limit of operation is reached.

When FIN received (LOW) → Make FOUT (LOW) → TIMER STARTS → TRP received (LOW) → TIMER STOPPED

After TRP received (LOW) then display/ output = TIME Between FOUT (LOW) and TRP (LOW)

TIME = 0.00 ms

COUNT OPERATION = + 1

Output Function

Developer is free to choose any of the following to display the values.

Values to Display / Output

I. TIME
II. COUNTER

Display Method

I. LCD or
II. 7 Segment Display or
III. RS232 Port

Source Code

- ✓ Require source code in Atmel Studio Format (with library if used)
- ✓ Developer will provide Fuse Bits to be used to program the Atmega128A MCU

If Developer is successful in running the above functions in the hardware then the SOW will be shared for complete Firmware Development and Project will be assigned after the final discussion on the SOW.